Report for lesson 5 Encoding work

This is report for lesson 5 but lets get remember previous lesson to understand content fully. First and foremost, almost all of the foundation lectures are going to start with one thing, and it is data preprocessing. Of course the person have to responsible and pick suitable and reliable source of information in order to avoid future difficulties but most important thing is preprocessing. The reason of the importance of the preprocessing lies on the usage of the dataset, cause algorithm face with difficulties and most of the time cause errors in the code which leads to both optimization and performance problems in the future. Also It is reminder for me to clear dataset and fill them carefully because while processing with dataset we can unconsciously fill them with synthetic information.

data preprocessing have certain specific ways to sort and clear information from dataset via performing with methods such as: mean, mode, median, fixed, and drop. Each of them have it's own cases for usage but still mode method can be universal way to fill gaps but is functionally just repeats the most used input feature and give it to all missing features. Today we are going to learn next step which is data encoding. Personally from this lecture I understand, that encoding mainly means transformation of the data types, for example from str to int or str to float, by manipulating type of the data we can sort dataset to more usable and understandable information to machine, which is mainly to assembly language I guess cause all of the code and algorithms are ends up their commands in assembly stage.

overall we have 5 types of encoding methods, but for now we started with 2 of them. Additionally there are also advanced types of data manipulation, and ways to fill up missing gaps in dataset but lets talk more about encoding. So first method of encoding is one-hot encoding method. Which means transformation of the information from str type to int or whatever by creating new categorized dataset column that also have specialized class in which machine will print result according to answer of the condition(like is this value equal to another or something like that) usually It is 1 or 0 answers. Although it requires more action to implementation like creating dummies, dropping old segment of dataset then concatenation of datasets between old dataset and dummy.

Second method is more easier than first one, it only needs special library which is scikit-learn library. We had to download it in terminal import it trough sklearn.preprocessing LabelEncoder. It only need few more commands like fit-transform with the segment of the dataset and it is done. Only difference between them is that first method creates new column of information each time when it faces new value(for example we have 20 types of vegetables, and if I try to do it with one-hot method it can process it of course but with each iteration it create columns where mostly information are going to look like 19 zeros and 1 one value. This way suitable for small indexed features while second more convenient for me personally)